Problem 20

Determine the limit to which the following expression tends as $n \to \infty$:

$$a_n = \frac{1}{n^{1+\alpha}} \sum_{k=1}^{n} k^\alpha \quad (\alpha > -1).$$

Solution

Step 1: Recognize as a Riemann Sum

We can rewrite the expression as:

$$a_n = \frac{1}{n^{1+\alpha}} \sum_{k=1}^{n} k^\alpha = \frac{1}{n} \sum_{k=1}^{n} \left(\frac{k}{n}\right)^\alpha.$$

This is a Riemann sum for the function $f(x) = x^\alpha$ on the interval $[0, 1]$, with partition points $x_k = \frac{k}{n}$ and subinterval width $\Delta x = \frac{1}{n}$.

Step 2: Convert to Riemann Integral

As $n \to \infty$, the Riemann sum converges to the definite integral:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \left(\frac{k}{n}\right)^\alpha = \int_0^1 x^\alpha \, dx.$$

Step 3: Evaluate the Integral

For $\alpha > -1$, the integral is well-defined:

$$\int_0^1 x^\alpha \, dx = \left[\frac{x^{\alpha+1}}{\alpha+1}\right]_0^1 = \frac{1^{\alpha+1}}{\alpha+1} - \frac{0^{\alpha+1}}{\alpha+1} = \frac{1}{\alpha+1}.$$

Note: The condition $\alpha > -1$ ensures that $\alpha + 1 > 0$, which makes the integral convergent at $x = 0$.

Step 4: State the Limit

Therefore:

$$\lim_{n\to\infty} a_n = \frac{1}{\alpha+1}.$$

Step 5: Verification Using Summation Formulas

We can verify this result for specific values of α:

Case 1: $\alpha = 0$

$$a_n = \frac{1}{n} \sum_{k=1}^{n} 1 = \frac{n}{n} = 1.$$

According to our formula: $\frac{1}{\alpha+1} = \frac{1}{0+1} = 1$.

Case 2: $\alpha = 1$

$$a_n = \frac{1}{n^2} \sum_{k=1}^{n} k = \frac{1}{n^2} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2n} = \frac{1}{2} + \frac{1}{2n}.$$

As $n \to \infty$: $a_n \to \frac{1}{2}$. According to our formula: $\frac{1}{\alpha+1} = \frac{1}{1+1} = \frac{1}{2}$.

Case 3: $\alpha = 2$

$$a_n = \frac{1}{n^3}\sum_{k=1}^{n} k^2 = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6}.$$

Simplifying:

$$a_n = \frac{n(n+1)(2n+1)}{6n^3} = \frac{(n+1)(2n+1)}{6n^2} = \frac{2n^2+3n+1}{6n^2} = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}.$$

As $n \to \infty$: $a_n \to \frac{1}{3}$. According to our formula: $\frac{1}{\alpha+1} = \frac{1}{2+1} = \frac{1}{3}$.

Step 6: Riemann Sum Interpretation

The key insight is recognizing that:

$$\frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right) \to \int_0^1 f(x)\,dx$$

as $n \to \infty$ for any Riemann integrable function f. This is the fundamental connection between discrete sums and continuous integrals, which is the foundation of integral calculus.

Final Answer

The limit of the expression is:

$$\boxed{\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{1}{n^{1+\alpha}}\sum_{k=1}^{n} k^\alpha = \frac{1}{\alpha+1} \quad \text{for } \alpha > -1.}$$

This result follows from interpreting the expression as a Riemann sum for $\int_0^1 x^\alpha\,dx$, which equals $\frac{1}{\alpha+1}$.